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Questions

1. Explain the setup of supervised learning

Intuition:

Supervised learning is a paradigm where we have a dataset with both input features and corresponding output labels. The goal is to learn a mapping from inputs to outputs so that we can predict the label for new, unseen inputs.

Detailed Answer:

In supervised learning, we are given a dataset $X = \{x_1, \dots, x_N\} \subset \Omega \subseteq \mathbb{R}^D$ and a set of labels $Y = \{y_1, \dots, y_N\} \subset \mathbb{N}$ (for classification) or $\mathbb{R}$ (for regression). Each pair (x_i, y_i) is an example, where x_i is the feature vector and y_i is the target label.

The objective is to find a function $\phi_\theta : \Omega \to Y$ (parameterized by θ) that minimizes a loss function $L_{X \times Y}(\theta)$ which measures the discrepancy between the predicted labels $\phi_\theta(x_i)$ and the true labels y_i. Common loss functions include mean squared error for regression and cross-entropy for classification.

The learning process involves adjusting the parameters θ to minimize the loss, typically using optimization algorithms like gradient descent. Once learned, ϕ_θ can be used to predict labels for new data points.

Examples of supervised learners include linear regression, logistic regression, neural networks, and support vector machines.

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2. Why is LDA a linear method?

Intuition:

LDA finds a linear combination of features that best separates the classes. The decision boundary in the transformed space is linear.

Detailed Answer:

LDA is a linear method because it seeks a projection direction (or directions) $a \in \mathbb{R}^D$ such that the projected data $\hat{x} = a^T x$ maximizes class separation. The criterion used (Fisher's criterion) leads to a solution that is a linear function of the input features.

Specifically, LDA finds a set of linear discriminants by solving a generalized eigenvalue problem $\Sigma_b a = \lambda \Sigma_w a$, where Σ_b and Σ_w are the between-class and within-class covariance matrices. The eigenvectors a define linear transformations of the input space.

Furthermore, when used for classification, LDA assumes that each class follows a Gaussian distribution with a common covariance matrix, leading to a linear decision boundary. The classification rule is based on linear functions of the input x.

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3. What characterizes a discriminant direction? What does a represent?

Intuition:

A discriminant direction is a direction in the feature space along which the projected data shows maximal separation between classes and minimal scatter within each class.

Detailed Answer:

A discriminant direction, represented by a vector $a \in \mathbb{R}^D$, is characterized by the property that when data points are projected onto this direction, the ratio of between-class variance to within-class variance is maximized.

Mathematically, for two classes, the projected means are $\hat{\mu}_1 = a^T \mu_1$ and $\hat{\mu}_2 = a^T \mu_2$, and the within-class variances are $\hat{\sigma}_1^2 = a^T \Sigma_1 a$ and $\hat{\sigma}_2^2 = a^T \Sigma_2 a$ (assuming common covariance Σ_w). Fisher's criterion seeks a that maximizes:

$$J(a) = \frac{(\hat{\mu}_1 - \hat{\mu}_2)^2}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2} = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}.$$

The vector a represents the direction in the original D-dimensional space that best separates the classes. It is the linear combination of the original features that yields the most discriminative feature.

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4. What is the Fisher criteria?

Intuition:

Fisher's criterion is a measure of class separability. It is defined as the ratio of the between-class variance to the within-class variance. Maximizing this ratio leads to optimal class separation.

Detailed Answer:

Fisher's criterion, for two classes, is defined as:

$$J(a) = \frac{(\hat{\mu}_1 - \hat{\mu}_2)^2}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2},$$

where $\hat{\mu}_k$ is the mean of the projections of class k, and $\hat{\sigma}_k^2$ is the variance of the projections of class k.

In the multi-class case, the criterion generalizes to:

$$J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a},$$

where Σ_b is the between-class covariance matrix and Σ_w is the within-class covariance matrix.

Maximizing $J(a)$ leads to a projection direction that maximizes the distance between class means (numerator) while minimizing the spread within each class (denominator). This is the fundamental objective of LDA.

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5. What is the inter-class (between) criteria? Explain its principle.

Intuition:

The inter-class (or between-class) criterion measures how far apart the class means are. A good discriminant direction should have the projected class means as far apart as possible.

Detailed Answer:

The inter-class criterion aims to maximize the separation between the class centroids after projection. For two classes, this is simply the squared distance between the projected means: $(\hat{\mu}_1 - \hat{\mu}_2)^2$.

For M classes, we define the between-class covariance matrix:

$$\Sigma_b = \frac{1}{M} \sum_{k=1}^{M} (\mu_k - \mu)(\mu_k - \mu)^T,$$

where μ_k is the mean of class k and μ is the global mean.

The between-class criterion for a projection direction a is then $a^T \Sigma_b a$. Maximizing this quantity encourages the projected class means to be spread out.

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6. How to compute the inter-class covariance matrix?

Intuition:

The inter-class covariance matrix is computed as the covariance of the class means around the global mean, weighted by the class sizes.

Detailed Answer:

Given M classes, let μ_k be the mean vector of class k, N_k be the number of samples in class k, and $\mu = \frac{1}{N} \sum_{k=1}^{M} N_k \mu_k$ be the global mean (with $N = \sum_k N_k$).

The between-class covariance matrix Σ_b is computed as:

$$\Sigma_b = \frac{1}{M} \sum_{k=1}^{M} (\mu_k - \mu)(\mu_k - \mu)^T.$$

Alternatively, some formulations use a weighted version:

$$\Sigma_b = \sum_{k=1}^{M} N_k(\mu_k - \mu)(\mu_k - \mu)^T.$$

In either case, Σ_b captures the scatter of the class means around the global mean.

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7. What is the intra-class (within) criteria? Explain its principle.

Intuition:

The intra-class (or within-class) criterion measures how tightly clustered the data points are within each class after projection. A good discriminant direction should have minimal scatter within each class.

Detailed Answer:

The intra-class criterion aims to minimize the variance within each class after projection. For two classes, this is the sum of the variances of the projected data in each class: $\hat{\sigma}_1^2 + \hat{\sigma}_2^2$.

For M classes, we define the within-class covariance matrix as:

$$\Sigma_w = \sum_{k=1}^{M} \Sigma_k,$$

where Σ_k is the covariance matrix of class k. Alternatively, if we assume a common covariance matrix, we can compute Σ_w as the pooled covariance:

$$\Sigma_w = \frac{1}{N} \sum_{k=1}^{M} \sum_{y_i = k} (x_i - \mu_k)(x_i - \mu_k)^T.$$

The within-class criterion for a projection direction a is $a^T \Sigma_w a$. Minimizing this quantity encourages the data points within each class to be tightly clustered around their class mean.

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8. How to compute the intra-class covariance matrix?

Intuition:

The intra-class covariance matrix is computed as the sum of the covariance matrices of each class, or as the average of the squared deviations of data points from their class mean.

Detailed Answer:

For each class k, compute the centered data matrix $A_k = [x_1^{(k)} - \mu_k, \dots, x_{N_k}^{(k)} - \mu_k]$, where $x_i^{(k)}$ are the data points of class k and μ_k is the class mean.

Then, the within-class covariance matrix for class k is $\frac{1}{N_k} A_k A_k^T$. The overall within-class covariance matrix is the sum (or average) of these class-specific matrices:

$$\Sigma_w = \sum_{k=1}^{M} \frac{1}{N_k} A_k A_k^T.$$

Alternatively, if we assume equal class sizes, we can compute Σ_w as:

$$\Sigma_w = \frac{1}{N} \sum_{k=1}^{M} \sum_{y_i=k} (x_i - \mu_k)(x_i - \mu_k)^T.$$

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9. How are they both combined?

Intuition:

Fisher's criterion combines the between-class and within-class criteria by taking their ratio. We maximize this ratio to find a direction that simultaneously maximizes class separation and minimizes within-class scatter.

Detailed Answer:

The combined criterion is Fisher's criterion:

$$J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}.$$

This ratio is known as the Rayleigh quotient. Maximizing $J(a)$ is equivalent to solving the generalized eigenvalue problem:

$$\Sigma_b a = \lambda \Sigma_w a.$$

The eigenvector a corresponding to the largest eigenvalue λ is the first discriminant direction. Subsequent directions are obtained by taking the eigenvectors corresponding to the next largest eigenvalues.

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10. Can you justify and explain the derivation of the maximum for a?

Intuition:

To maximize the ratio $J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$, we set the derivative with respect to a to zero. This leads to a generalized eigenvalue problem.

Detailed Answer:

We want to maximize $J(a) = \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$. Since $J(a)$ is scale-invariant (multiplying a by a scalar does not change the ratio), we can constrain $a^T \Sigma_w a = 1$ and maximize $a^T \Sigma_b a$.

Using Lagrange multipliers, we define the Lagrangian:

$$\mathcal{L}(a, \lambda) = a^T \Sigma_b a - \lambda(a^T \Sigma_w a - 1).$$

Taking the derivative with respect to a and setting to zero:

$$\frac{\partial \mathcal{L}}{\partial a} = 2\Sigma_b a - 2\lambda \Sigma_w a = 0 \quad \Rightarrow \quad \Sigma_b a = \lambda \Sigma_w a.$$

This is a generalized eigenvalue problem. The eigenvectors a that satisfy this equation are the discriminant directions, and the eigenvalues λ give the value of $J(a)$.

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11. Can you describe the multidimensional situation (D $>$ 2)?

Intuition:

In multidimensional space (D $>$ 2), LDA seeks multiple discriminant directions that form a subspace. The number of such directions is at most the number of classes minus one.

Detailed Answer:

For M classes in a D-dimensional space (with $D > 2$), LDA seeks a set of discriminant directions $a_1, a_2, \dots, a_{M-1}$ that form an orthogonal basis for a subspace of dimension at most $M - 1$.

We solve the generalized eigenvalue problem $\Sigma_b a = \lambda \Sigma_w a$ and take the eigenvectors corresponding to the largest eigenvalues. These eigenvectors are ordered by their discriminative power (the eigenvalue indicates the value of Fisher's criterion).

The data can then be projected onto this subspace for visualization or classification. Note that if $D > M$, there are at most $M - 1$ non-zero eigenvalues because the between-class covariance matrix Σ_b has rank at most $M - 1$.

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12. How to do inference using LDA?

Intuition:

Inference in LDA involves assigning a new data point to the class that maximizes the posterior probability, assuming Gaussian class-conditional distributions with a common covariance matrix.

Detailed Answer:

After obtaining the discriminant directions, we can use LDA for classification. The method assumes that each class k follows a Gaussian distribution $\mathcal{N}(\mu_k, \Sigma_w)$ (common covariance matrix).

For a new data point x, we compute the posterior probability $\mathbb{P}(C = k|x)$ using Bayes' rule:

$$\mathbb{P}(C = k|x) = \frac{\mathbb{P}(x|C = k)\mathbb{P}(C = k)}{\mathbb{P}(x)}.$$

Ignoring the constant evidence term, we assign x to the class k that maximizes the discriminant function:

$$\delta_k(x) = x^T \Sigma_w^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma_w^{-1} \mu_k + \log \mathbb{P}(C = k).$$

In practice, we often use the projected data in the discriminant subspace and perform classification using linear decision boundaries.

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13. Provide an example of conditional data model using LDA.

Intuition:

LDA can be seen as a generative model where each class is modeled by a Gaussian distribution with the same covariance matrix. The conditional distribution of the data given the class is Gaussian.

Detailed Answer:

Consider a classification problem with M classes. The conditional data model for LDA assumes:

$$x|C = k \sim \mathcal{N}(\mu_k, \Sigma_w),$$

where μ_k is the mean of class k and Σ_w is the common within-class covariance matrix.

The prior distribution over classes is $\mathbb{P}(C = k) = \pi_k$ (often estimated from the class frequencies).

Then, the joint distribution of x and C is:

$$\mathbb{P}(x, C = k) = \mathbb{P}(x|C = k)\mathbb{P}(C = k) = \frac{1}{(2\pi)^{D/2}|\Sigma_w|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \Sigma_w^{-1}(x - \mu_k)\right) \pi_k.$$

This model leads to linear decision boundaries. The classification rule is to assign x to the class k that maximizes the posterior $\mathbb{P}(C = k|x)$, which is equivalent to maximizing the discriminant function $\delta_k(x)$ defined above.
